Stock Market Analysis in SQL

Six NSE stocks, 1 Jan 2015 to 31 Jul 2018 (889 trading days each)

1. What this project is about

The project uses SQL to study the price history of six Indian stocks: Bajaj Auto, Eicher Motors, Hero Motocorp, TVS Motors, Infosys and TCS. The question is whether a classic trend-following rule, the **golden cross**, would have given useful Buy and Sell signals. A Buy is when the 20-day moving average of the closing price crosses above the 50-day average, and a Sell is the reverse. The project also asks whether the data can be trusted. It cannot be, without checking: two stocks contain a price halving that looks like a crash but is a bonus issue, and that changes the answers.

2. What I did and which tools I used

Step	What was done
Explore	Row counts, date range, top closes, yearly averages, and a check for missing values across all six tables (Tasks 1–4).
Build	20/50-day moving averages with window functions, a master table of all six closing prices, and golden-cross Buy/Sell/Hold signals using LAG (Tasks 5–9).
Scale	One query for all six stocks using chained CTEs and PARTITION BY, giving Buy/Sell counts and the latest signal per stock (Task 10).
Question	First-versus-last price change, worst single day per stock, then adjustment of TCS and Infosys prices for 1:1 bonus issues and a re-run of signals (Tasks 11–14).

Tools: SQL written for MySQL 8 (window functions, CTEs, LAG, ROW_NUMBER, CASE, UNION ALL, JOINs); the same queries were run in SQLite to test them. I cross-checked every result with an independent calculation in Python (pandas), and made the charts with matplotlib. I also compared each result with the checkpoints in the student guide and the numbers in the course deck.

3. Key SQL code

The full script is in stock_market_analysis.sql. These are the queries that matter most.

3.1 Moving averages (Task 5). A 20-day window is 19 PRECEDING rows plus the current one, and the CASE blanks the first 19 rows, which do not have a full window.

```
CREATE TABLE bajaj1 AS
SELECT date, close_price,
  CASE WHEN ROW_NUMBER() OVER (ORDER BY date) >= 20
       THEN ROUND(AVG(close_price) OVER (ORDER BY date
            ROWS BETWEEN 19 PRECEDING AND CURRENT ROW), 2) END AS ma20,
  CASE WHEN ROW_NUMBER() OVER (ORDER BY date) >= 50
       THEN ROUND(AVG(close_price) OVER (ORDER BY date
            ROWS BETWEEN 49 PRECEDING AND CURRENT ROW), 2) END AS ma50
FROM bajaj_auto;
```

3.2 Golden-cross signals (Task 7). A cross is a change between yesterday and today, so LAG supplies yesterday's averages; the NULL guard comes first.

```
CREATE TABLE bajaj2 AS
WITH t AS (
  SELECT date, close_price, ma20, ma50,
         LAG(ma20) OVER (ORDER BY date) AS prev_ma20,
         LAG(ma50) OVER (ORDER BY date) AS prev_ma50
  FROM bajaj1)
SELECT date, close_price,
  CASE WHEN ma20 IS NULL OR ma50 IS NULL OR prev_ma20 IS NULL
            OR prev_ma50 IS NULL                          THEN 'Hold'
       WHEN ma20 > ma50 AND prev_ma20 <= prev_ma50       THEN 'Buy'
       WHEN ma20 < ma50 AND prev_ma20 >= prev_ma50       THEN 'Sell'
       ELSE 'Hold' END AS `signal`
FROM t;
```

3.3 All six stocks in one query (Task 10). PARTITION BY stock restarts every window for each stock; ROW_NUMBER() ... DESC picks the latest signal. The ma, lagged and sig steps repeat the logic above with PARTITION BY (shortened here).

```
WITH prices AS (SELECT 'Bajaj Auto' AS stock, date, close_price FROM bajaj_auto
                UNION ALL SELECT 'Eicher Motors', date, close_price FROM eicher_motors
                UNION ALL ... /* Hero, Infosys, TCS, TVS */),
ma AS     (SELECT stock, date, close_price,
             CASE WHEN ROW_NUMBER() OVER (PARTITION BY stock ORDER BY date) >= 20
                  THEN AVG(close_price) OVER (PARTITION BY stock ORDER BY date
                       ROWS BETWEEN 19 PRECEDING AND CURRENT ROW) END AS ma20,
             ... /* ma50 the same way */ FROM prices),
lagged AS (SELECT *, LAG(ma20) OVER (PARTITION BY stock ORDER BY date) AS prev_ma20,
                     LAG(ma50) OVER (PARTITION BY stock ORDER BY date) AS prev_ma50 FROM ma),
sig AS    (SELECT stock, date, CASE ... END AS `signal` FROM lagged),   -- as in 3.2
latest AS (SELECT stock, date, `signal`,
                  ROW_NUMBER() OVER (PARTITION BY stock ORDER BY date DESC) AS rn
           FROM sig WHERE `signal` <> 'Hold')
SELECT s.stock, SUM(s.`signal`='Buy') AS buys, SUM(s.`signal`='Sell') AS sells,
       l.date AS last_signal_date, l.`signal` AS last_signal
FROM sig s JOIN latest l ON l.stock = s.stock AND l.rn = 1
GROUP BY s.stock, l.date, l.`signal` ORDER BY s.stock;
```

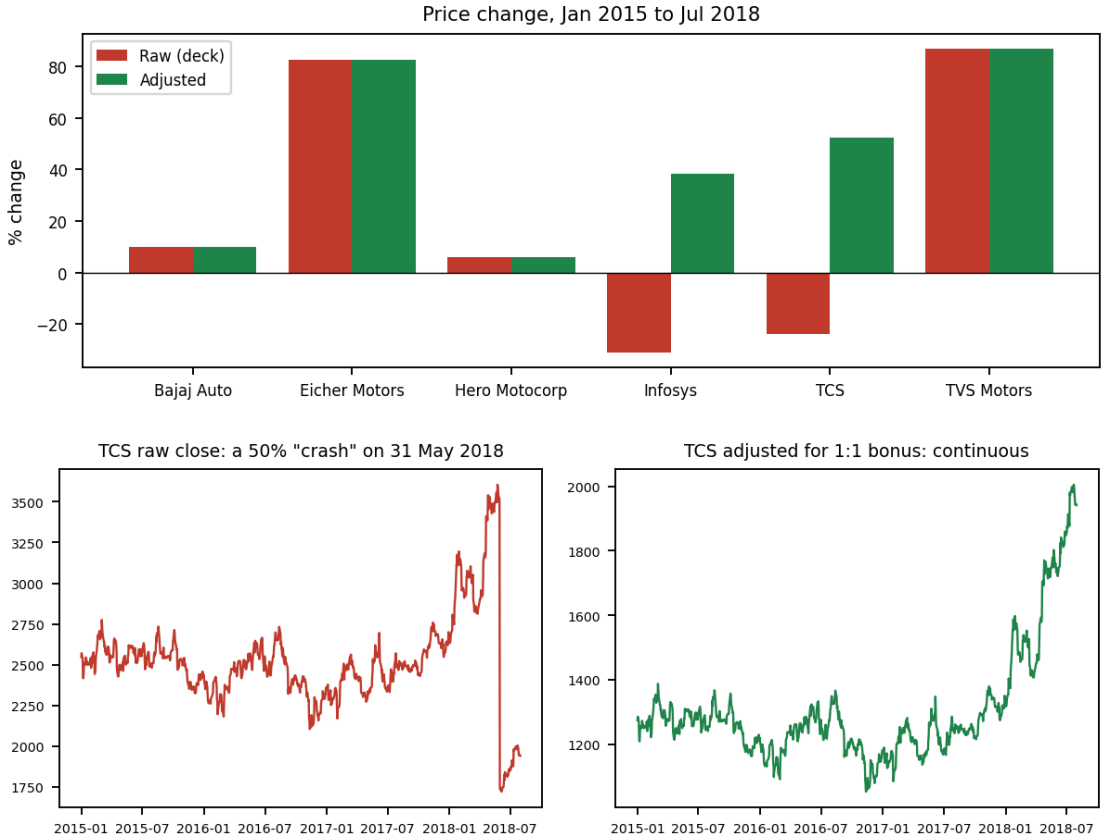
3.4 Finding the data trap (Task 12) and fixing it (Task 13). The worst-day query exposes two moves of about –50%; dividing earlier prices by 2 removes the cliff.

```
moves AS (SELECT stock, date, close_price,
            100.0 * (close_price / LAG(close_price) OVER (PARTITION BY stock ORDER BY date) - 1) AS pct_move
          FROM prices),
ranked AS (SELECT *, ROW_NUMBER() OVER (PARTITION BY stock ORDER BY pct_move) AS rn
           FROM moves WHERE pct_move IS NOT NULL)
SELECT stock, date, close_price, ROUND(pct_move,1) FROM ranked WHERE rn = 1;

-- adjustment for 1:1 bonus issues (TCS 2018-05-31, Infosys 2015-06-15)
CASE WHEN stock='TCS'     AND date < '2018-05-31' THEN close_price / 2
     WHEN stock='Infosys' AND date < '2015-06-15' THEN close_price / 2
     ELSE close_price END AS adj_close
```

4. Analysis and insights

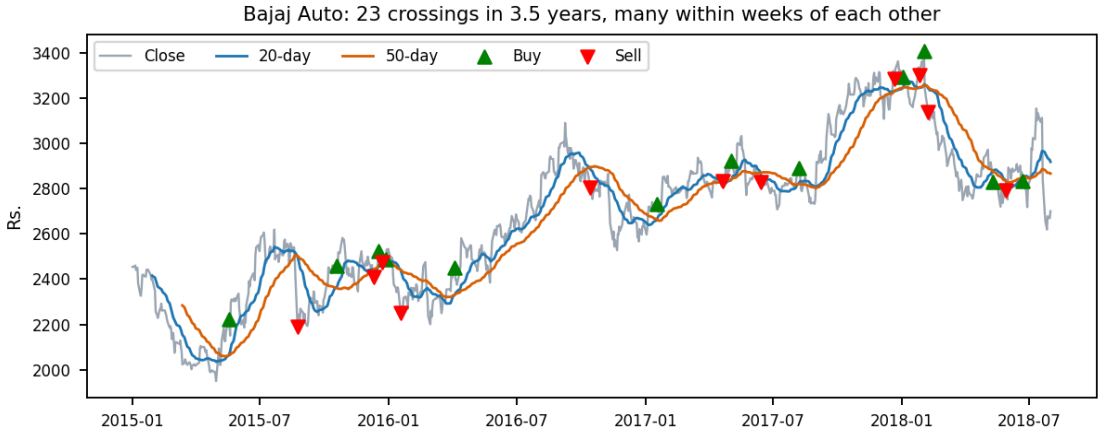
Headline. The course deck is wrong for TCS and Infosys. Both had a 1:1 bonus issue (Infosys on 15 Jun 2015, TCS on 31 May 2018): the price halved overnight with no loss of value. Raw prices show losses of –23.8% and –30.9%; after adjustment they are gains of +52.4% and +38.2%. This also removes the TCS Sell of 5 Jun 2018 and flips TCS's latest signal from Sell to Buy.



Evidence: Infosys fell 49.9% in one day (1,976.65 to 991.10) and TCS fell 50.4% (3,517.75 to 1,744.80). The other four stocks' worst days are between –6% and –10%. Caveat: I inferred the events from the price pattern; confirm against exchange corporate-action records.

Stock	Change (adj.)	Buys / Sells	Latest signal	20d vs 50d on last day	Signal strategy vs holding
Bajaj Auto	+10.0%	12 / 11	Buy, 21 Jun 2018	2,918 vs 2,867 (above)	-2.7% vs +10.0% (4 of 12 won)
Eicher Motors	+82.6%	6 / 7	Sell, 6 Jun 2018	27,757 vs 28,792 (below)	+76.5% vs +82.6% (6 of 6 won)
Hero Motocorp	+6.0%	9 / 9	Sell, 22 May 2018	3,388 vs 3,512 (below)	-16.3% vs +6.0% (3 of 9 won)
Infosys	+38.2%	10 / 10	Buy, 7 May 2018	1,334 vs 1,286 (above)	+7.1% vs +38.2% (4 of 10 won)
TCS	+52.4%	12 / 12	Buy, 20 Apr 2018	1,955 vs 1,859 (above)	-22.4% vs +52.4% (3 of 12 won)
TVS Motors	+86.9%	8 / 8	Sell, 17 May 2018	560 vs 570 (below)	+85.0% vs +86.9% (3 of 8 won)

Strategy: buy at the close of each Buy signal, sell at the next Sell, hold cash in between; open trade valued at the last close; costs ignored (roughly 0.1% each way would cut each figure by about 1.5 to 3 points).



Whipsaws. The golden cross lagged badly in sideways markets. Bajaj Auto had 23 signals, 11 of them within 30 days of the previous one; only 4 of its 12 round trips made money, so following the signals returned –2.7% against +10.0% for holding. Hero won 3 of 9 trades (–16.3% against +6.0%). TCS won 3 of 12 (–22.4% against +52.4%). Eicher is the cleanest trend-follower: all 6 trades won, and the strategy returned +76.5% against +82.6% for holding.

Does the latest signal agree with the long-run trend? It does for Bajaj, Infosys and TCS. For Eicher and TVS (both up more than 80%) the latest signal is a Sell, which follows a big run and so signals a pullback, not a reversal; for those I would trust the long-run trend over the signal. Hero's Sell fits a stock that has barely moved (+6.0%).

Method caveats. A moving average uses only past prices, so a golden cross needs 50 days of history and confirms a move after it has happened (Bajaj's first Sell, 24 Aug 2015, fired on the same day as its worst one-day fall, –9.1%). Missing from the data: dividends, news and earnings, brokerage and taxes, the market index, and a longer history; four of the six stocks are auto companies, so results are correlated. The only data gap is a single missing deliverable-quantity value per stock (9 Dec 2015 and 31 Aug 2017), which does not affect any price result.

5. Recommendations

These follow from this data only and are not investment advice; they ignore dividends, costs, valuation and the wider market.

Item	Recommendation	Why
TCS, Infosys	Treat as positive performers; drop the deck's "sell TCS" call.	Adjusted gains of +52.4% and +38.2%; both latest signals are Buy.
Eicher, TVS	Strong long-run performers; use the Sell signals as a caution to watch, not a reason to exit on their own.	+82.6% and +86.9%; the Sells follow large runs and the averages are still close.
Bajaj Auto	No clear edge from the signals; wait for confirmation beyond a single crossover.	+10.0% with 23 signals, 11 within a month of another.
Hero Motocorp	Weakest case: low return and a Sell signal.	+6.0% over the period and the strategy lost 16.3%.
Method	Adjust prices for corporate actions before computing returns or signals, and add a confirmation filter (for example a minimum gap between signals) before relying on a golden cross.	The unadjusted data gave a wrong answer for 2 of 6 stocks; the signal strategy trailed buy-and-hold on all six stocks.